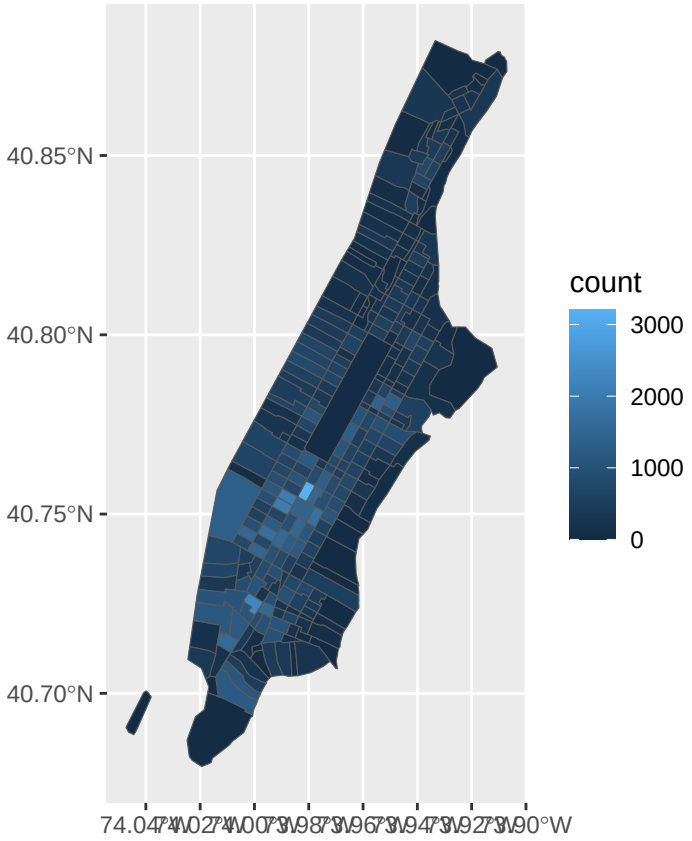
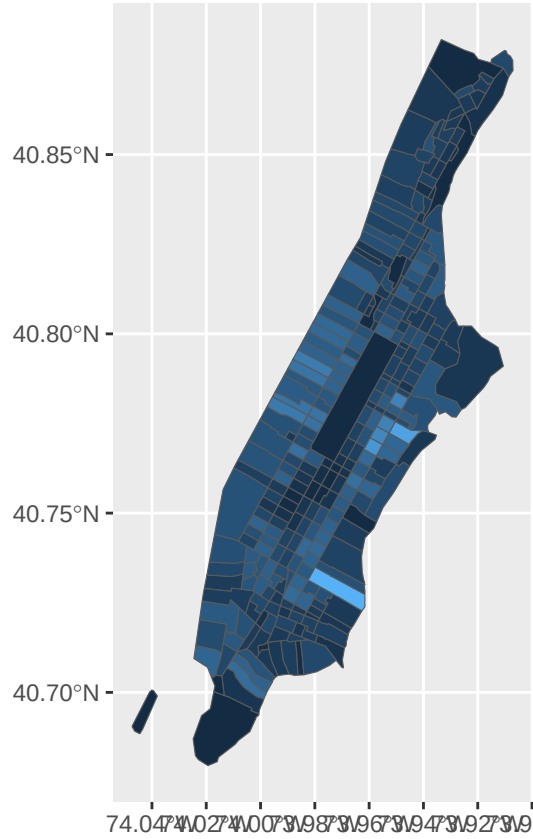
Untitled

2024-01-10



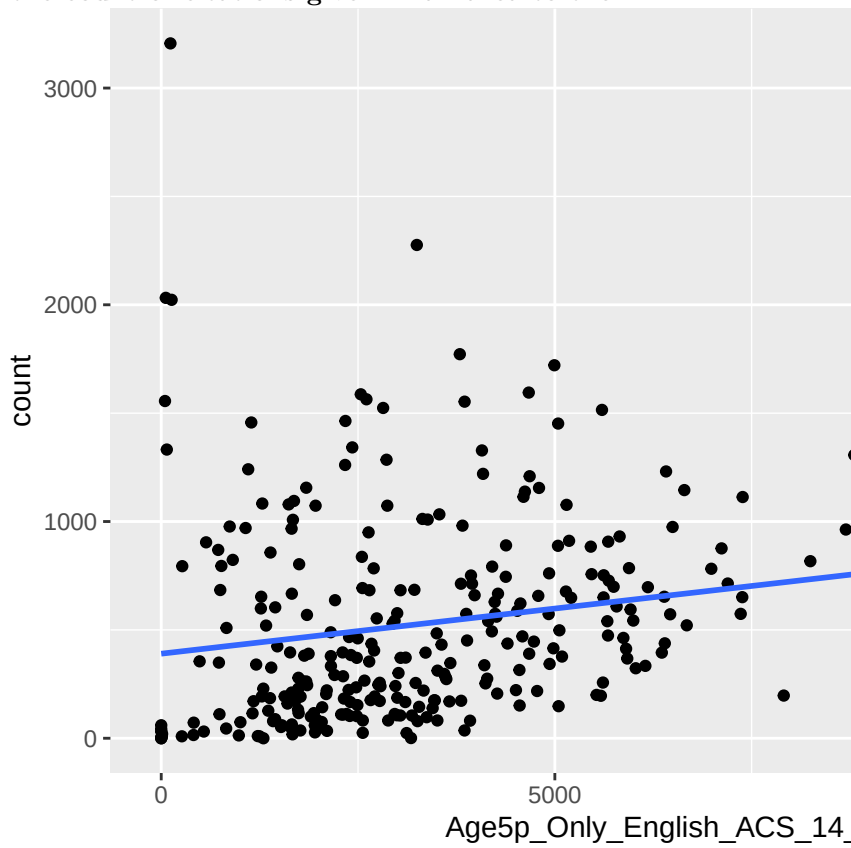

```
## [1] "18739e63ce165a3645bdfd030fd6eccbb65ca8b4"
```

English Level



Heat map showing the income English level in sections of Manhattan

Scatter plot and Regression Line comparing the count of citations given in an area to the income English level in sections of Manhattan



Ascending regression line shows that as the number of persons who speak English on the x axis increases the number of citations given in the area also increase. This is a major contrast to the hypothesis stated earlier.

```
## # A tibble: 2 x 7
##   term                    estimate std_error statistic p_value lower_ci upper_ci
##   <chr>                      <dbl>     <dbl>     <dbl>   <dbl>    <dbl>    <dbl>
## 1 intercept                 390.       50.2        7.77   0      291.     489.
## 2 Age5p_Only_English_ACS~    0.042      0.013      3.20   0.002    0.016    0.067
```

Based on the graph and the regression line created we see that when the number of persons speaking English is zero (0.042) the number of citation given is 390.22.

$0.042 = 42/1000$
For every 1000 people 42 people speak English well.
When the number of people who speak English in an area goes up by one the count changes on average of 42 people.

```
## Simple feature collection with 1 feature and 1 field
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: -74.04731 ymin: 40.67957 xmax: -73.907 ymax: 40.88207
## Geodetic CRS:  NAD83
##   cor(Age5p_Only_English_ACS_14_18, count)                       geometry
## 1                                 0.1861536 MULTIPOLYGON (((-74.00985 4...
```

Creating a shape file to plot areas where citations have been given.

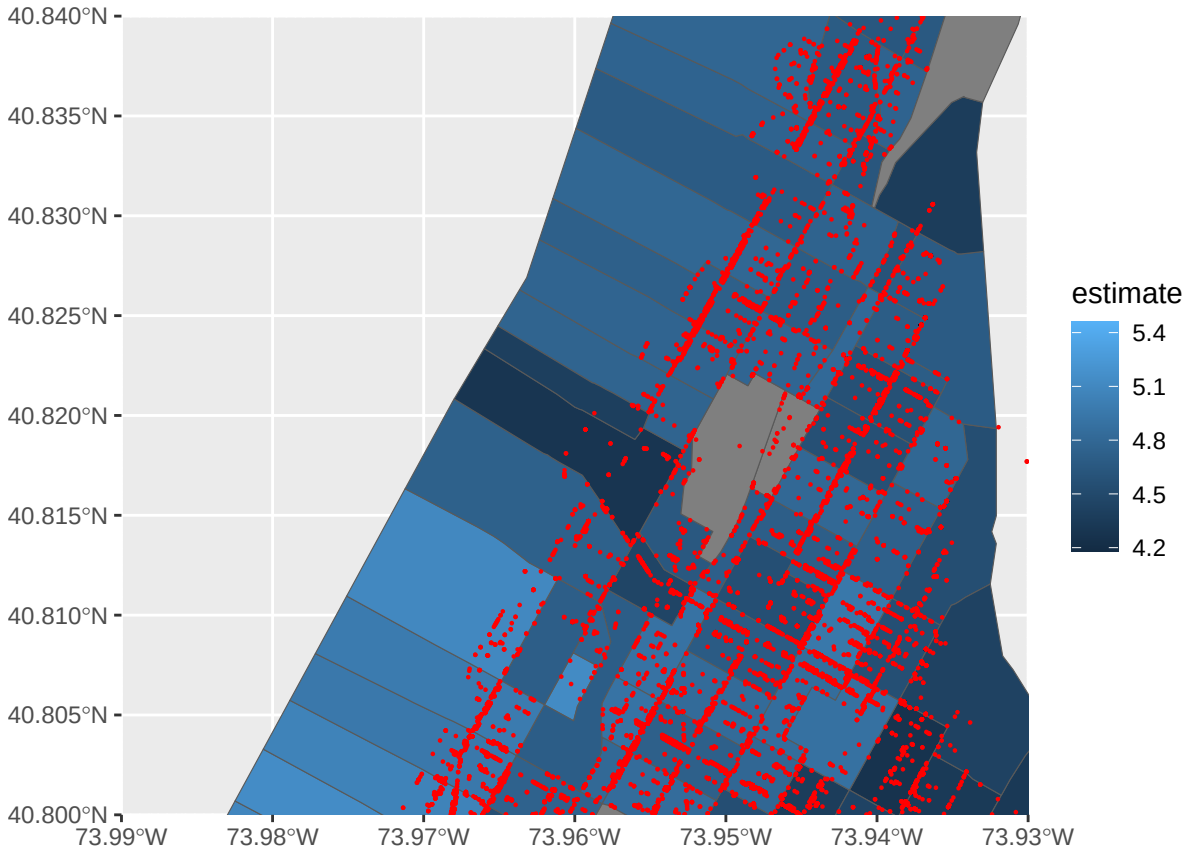
EasyCensus data

EasyCensus lets us search for useful census variables.

The variable `B19081` looks useful. We can get this information for NY with the following function.

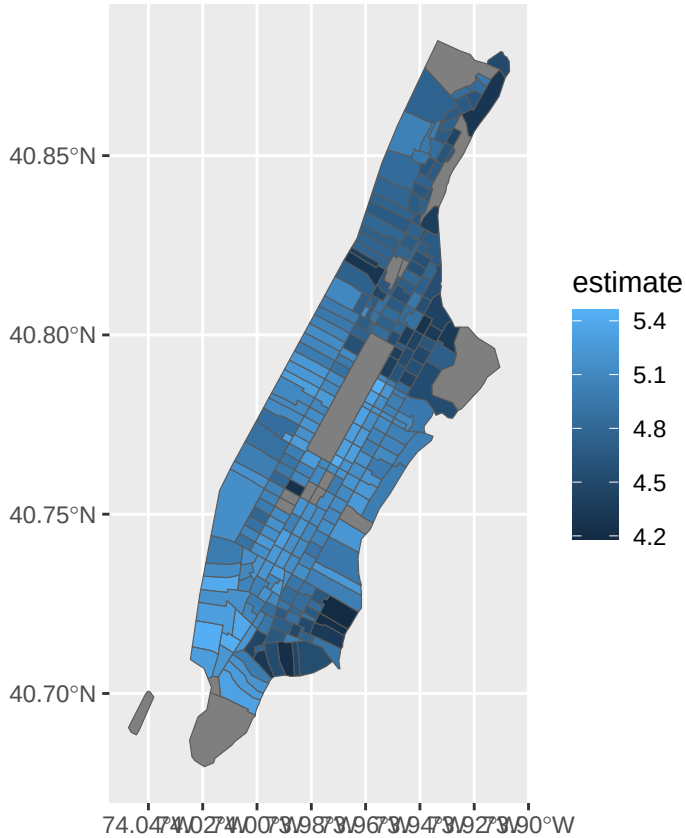
We should join this data to our shape file.

Now we can make a plot of this:



Income Level

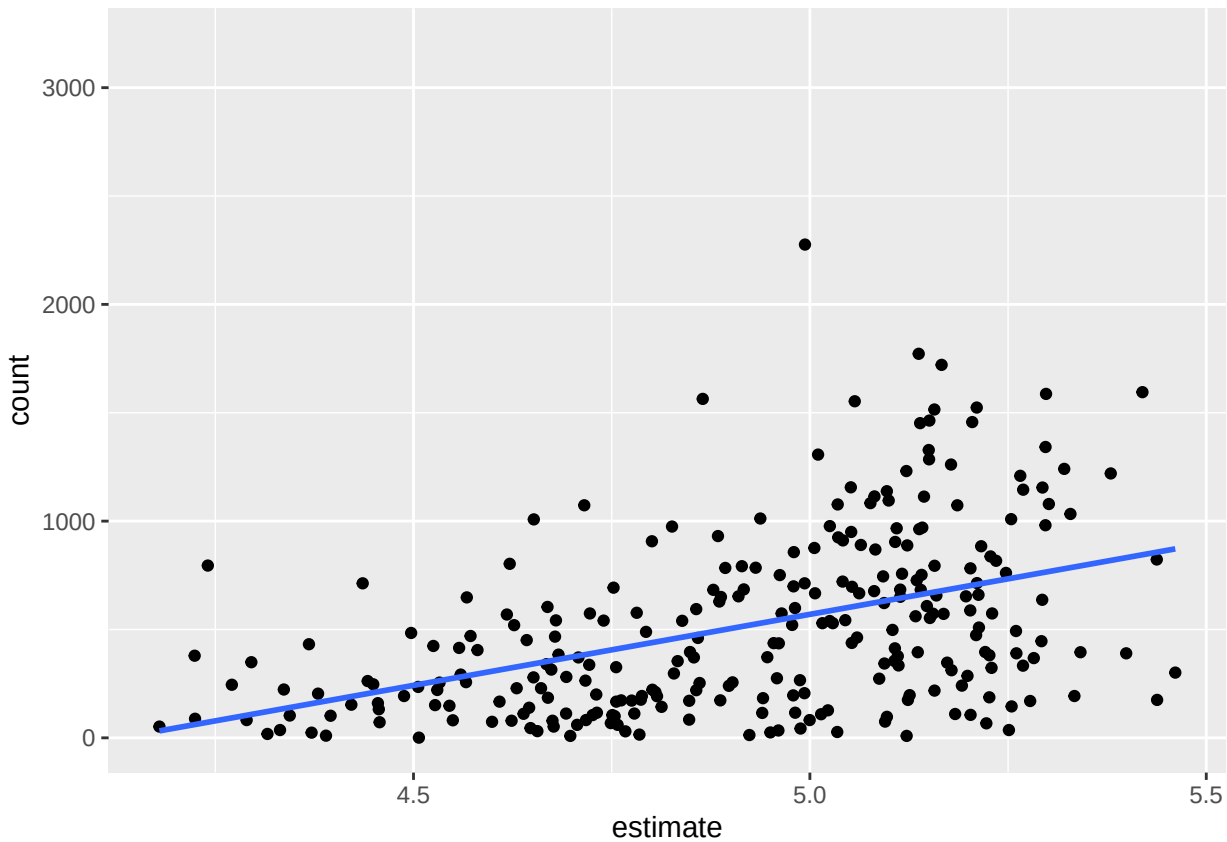
Heat map showing the income quintile level in sections of Manhattan (Hundred Thousand Dollars)



Scatter plot and Regression Line comparing the count of citations given in an area to the income quintile level in sections of Manhattan (Hundred Thousand Dollars)

```
## Warning: Removed 15 rows containing non-finite values (`stat_smooth()`).
```

```
## Warning: Removed 15 rows containing missing values (`geom_point()`).
```



Ascending regression line shows that as the number of persons who makes more income on the x axis increases the number of citations given in the area also increase. This is a major contrast to the hypothesis stated earlier.

```
## # A tibble: 2 x 7
##   term      estimate std_error statistic p_value lower_ci upper_ci
##   <chr>        <dbl>     <dbl>     <dbl>   <dbl>    <dbl>    <dbl>
## 1 intercept   -2707.     391.      -6.92       0   -3477.   -1937.
## 2 estimate      655.      79.4      8.25       0     499.     812.
```

Based on the graph and the regression line created we see that when the number of persons with an income of zero (655.364) the number of citation given is -2706.957

655.364/1000 = 655,364
For every 1000 people there is a change in income of $655,364 .
When the income level each person has in the area goes up by one the count changes on average of ( ) people.